Australian Wildlife Classifier – AWC135

User Guide

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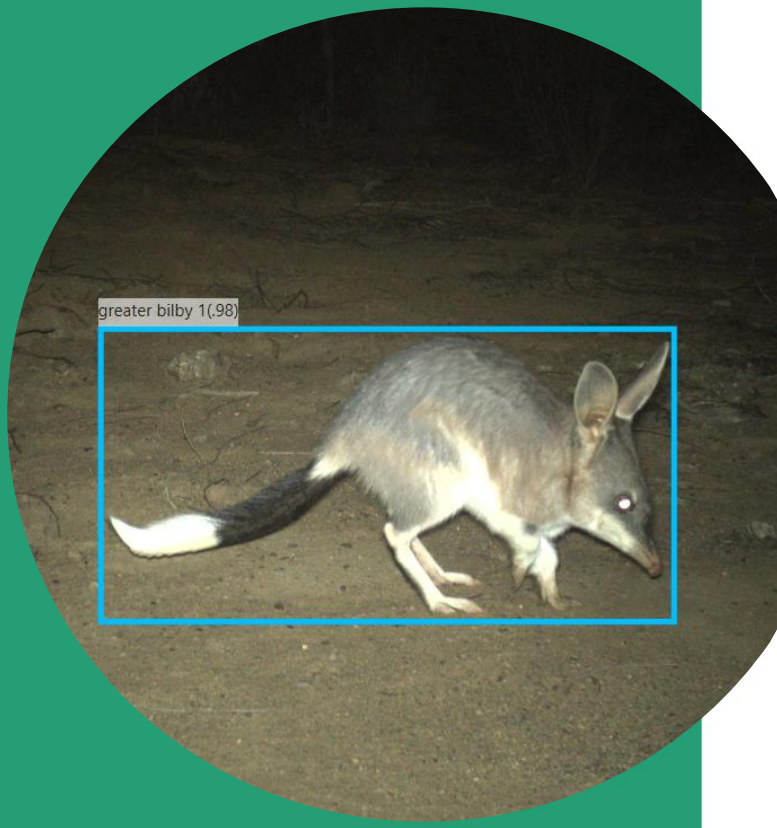

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Background information

Image recognition: detection vs classification

Image recognition is a function of machine learning that enables AI to interpret and interact with visual data such as motion-sensor camera images. The first level of this is object detection, which is recognising objects within an image and identifying them visually with a bounding box. An object detection model is an AI model with the task of taking a raw image and pointing out the shapes that are important, based on what it has been trained to recognise. One of the most well-known models of this kind is the MegaDetector, which is an open-source object detection model that is trained to recognise and identify three labels: 'animal', 'person, and vehicle. This output is used as the base information for our classification model, AWC135.

The second level of image recognition is image classification. This is the process of labelling images into a set of trained categories. To establish these categories, a model is given training images and told which image fits which category. These categories are given labels that make sense for the data, such as 'domestic cat' and 'quoll sp'. The model then must learn which features are consistent within the images of the same label, and which features are different between images of different labels. Our model, AWC135, uses the Megadetector's output of "animal" labels as a filter to exclude empty images. It also uses the bounding box locations to crop the image and allow the AI to focus on only the features of the animal and its immediate surround background.

Training images

The quality and breadth of images used to train a model directly influences what features and parameters it uses to classify images. The classic example is a model that was trained to identify whether a canine was a wolf or a husky, which performed well generally but misidentified some very obvious images. The answer was the training dataset, where all the images of wolves contained snow in the background, while images of huskies did not have snow in the background. This led to the model prioritising snow in the image as a top parameter for classification; detection of snow meant classification of wolf, even when the canine was clearly a husky.

The training images used for model AWC135 contain as much variety per species as possible. For most species, this includes a spread of images at different camera angles, from different places across the country, across infrared and white flash cameras. However, there are some limitations. A few species have restricted ranges, and thus were only captured on one model of camera, at one site, and with one orientational set up. Despite our best efforts, there are still some training image sets that contain only colour or only black and white images. We

have included a table in appendix 5 which contains a brief description of factors relating to camera set up and background context for each label. It is important to understand that the training data contributes to the parameters for classification, and how this may impact the model's performance on your dataset. There are situations where model AWC135 may misidentify the species in a dataset, but it is very likely to still be useful for aggregating and filtering visually similar species.

Confidence values

For every detection and classification that the AI models make, there will be associated confidence values for the label. The detection confidence is related to the object detection model's (MegaDetector) confidence that it has correctly identified an object in the image. The classification model assumes each detection is correct, so its confidence value is only related to how certain the model is in its classification of that object.

The classification label confidence value is the first number immediately beside the label. The number in brackets refers to the object detection confidence. The order is always: **classification label, classification confidence value (detection confidence value)**.

Confidence values quantify the model's confidence in its label prediction, and are not directly proportionate to the rate or probability of accuracy of the detection or classification. The value is a representation of how closely the image fits the parameters defined by the training data. Depending on aspects of an image and the training data, the image may be misclassified with high confidence if the model is picking up information that closely fits a parameter for a particular label, but that information is unrelated to the animal itself (i.e. background context), or defined by a bias in the training data for that label (e.g. higher representation of black and white images for one species over another).

How closely your images resemble the training images will greatly impact the confidence of AWC135's classifications. As such, it is not possible to recommend exact confidence values to use for filtering. One bounding box can have confidence values for multiple species predictions. The values are standardised to a maximum total of 1.0 per bounding box, with only the highest value being displayed. Below are some suggested ranges of confidence values and what is likely to be returned from a search in that range. While explicit confidence values cannot be offered, these suggested ranges will assist first time users with a starting place for utilising the recognition results. It will take some playing around with your data to figure out what confidence values to use for different searches.

Very high	0.95–1.00	Very high confidence searches are restrictive and will usually return a small proportion of images. However, the proportion of misidentified images will be very low. Images will be clear and often full body views of the animals. It will be easy to flick through and notice any incorrect identifications.
High	0.80–1.00	High confidence searches are less restrictive, so expect to view the bulk of the clear/easy to identify images. It will usually be obvious why the AI has predicted the species label, whether it is correct or not.
Moderate	0.50–0.80	Moderate confidence searches will return images that are difficult for the AI to identify, usually because the animal is only partially in view. There will still be a high number of correctly identified images, however, this is the range where visually similar species may get incorrectly identified and false positive objects (where the MegaDetector has picked up something inanimate as an animal) are classified.
Low	0.30–0.50	Low confidence searches will include false positive results and poor quality images of animals where the key identifying characteristics are not visible. Correctly identified images will still appear in this range, however, it may be difficult to verify them with the images out of context of their trigger sequence.

These confidence ranges do not apply to searches using the detection confidence (i.e. searching for any animal within a certain confidence range). The MegaDetector has its own range of suggested confidence thresholds based on the model version in use. For the MegaDetector v5, the suggested confidence threshold is around 0.20, where you can reasonably expect detections below this confidence value to not be animals. For the more recent version, v1000 redwood, that threshold has not yet been determined. It is likely to sit around a similar level, though this will require some exploration on the rate of correct and incorrect detections at different confidence values in your datasets. The simplest assumption to get you started is that images with no detections can be excluded.

Limitations

It is important to understand the constraints associated with the AI models. The following limitations can impact the accuracy of results within datasets, and should be considered when interpreting the results.

Training data limitations

As discussed above, the biggest limitation can be the training data, which defines the parameters and rules that are considered when predicting a label for a new image. Therefore, any biases, mistakes, and poor quality data can influence the AI to create rules that hinder its accuracy in predicting correct labels in real world datasets. Essentially, any highly repeated element across the training set will be noticed and identified as an important factor to consider. Examples of some of these elements are: camera angle; colour/black and white images; background context like rocks or grass; and non-target objects like bait containers. Our species verification process included the removal of poor quality images to reduce the influence of these elements. However, the variability of training images was reduced for rare and spatially restricted species.

Limitations of current AI framework

The following limitations are inherent to the current AI modelling framework. The training images define the parameters for each label based on the visual information only. Taxonomic relatedness among species and their geographic ranges are not considered.

Once trained, the model processes images in batches, treating each image as an independent entity. Although images are usually captured in a burst of 3–5 per trigger event, the models cannot incorporate that temporal relatedness into the label prediction. Even multiple bounding boxes within the same image are viewed and classified in isolation.

The species classifier can only label bounding boxes that are identified by the object detection model, so occasionally animals are missed at the detection level. More often though, non-animal objects are incorrectly identified as animals, which forces the species classifier to assign a species label to an inanimate object.

Species predictions are based solely on visual similarity to training images. The model does not work hierarchically within taxonomic levels, and label predictions cannot be influenced by the spatial location of new images. Additionally, the models do not estimate what proportion of all possible species labels is represented in its predictions. Model AWC135 calculates the confidence of its label prediction under the assumption that its 135 labels are all possible species options. These limitations are unfortunately a factor of the foundation of AI models at this point in time.

Interpretation of outputs

A further limitation relates to how the results are interpreted. It is easy to assume that the AI can replace human review and spit out an excel spreadsheet of results with a predefined error rate, however, this is not realistic. Motion-sensor camera images often contain animals that are only partially visible, moving, distant or poorly lit. Since correct label prediction

depends on the visibility of the animals' diagnostic characteristics, it is not feasible for the AI to label all the images without human verification.

Another common misunderstanding arises from confidence values. The model may produce moderate confidence values for correctly labelled species and occasional high confidence mistakes. These obvious mistakes draw disproportionate attention and lead to the conclusion that the labels cannot be used. As many species are not included in the model AWC135, there will be times where the species label is incorrect. However, since the model is labelled based on visual similarity to information and species that it is trained on, it will likely still label unrepresented species in a consistent way. This consistency can be more useful than strict taxonomic accuracy. So, please take into consideration that incorrect species labels can still aggregate visually similar images of species, supporting more efficient review and processing.

Australian Wildlife Classifier — AWC135

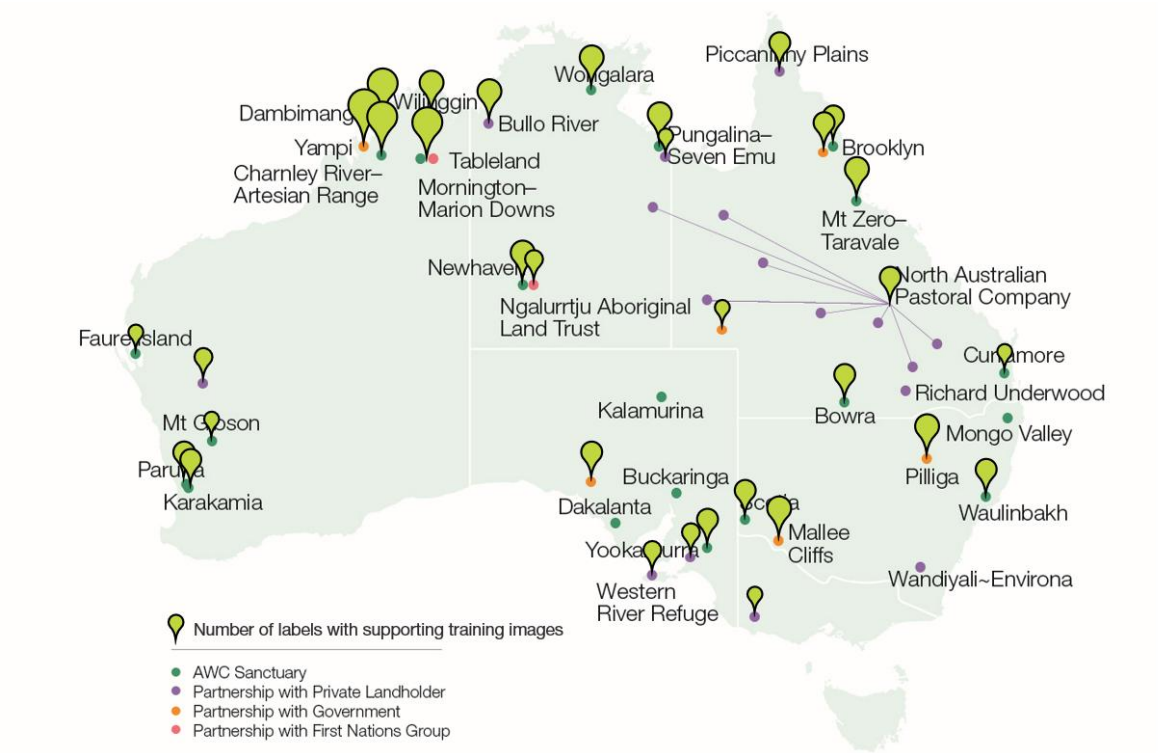
Model AWC135 was designed to increase the efficiency of processing images from motion-sensor cameras by creating an Australian model that contains a diverse range of common, threatened, and invasive species. With the spread of species included, this model should fit most use cases, including both general biodiversity monitoring and targeted surveys.

AWC135 classifies 135 labels of Australian species from 33 different source locations across the country. Several grouped labels were created with images from two or more species for either visually similar animals (e.g. 'grey kangaroo sp' contains images of both Eastern and Western Grey Kangaroos), or low importance/low abundance species (e.g. 'wetland bird sp' contains images of 15 different waterbird species such as Herons, Ibis, and Bush-hens). As a result of these grouped labels, images from over 200 species are included across the 135 labels. Some of those species contributed a very small number of images, so the threatened status of only the top 170 are included in the table below.

Nationally listed: Critically Endangered	4
Nationally listed: Endangered	16
Nationally listed: Endangered/Vulnerable	2
Nationally listed: Vulnerable	14
State listed: Critically Endangered	3
State listed: Endangered	12
State listed: Vulnerable	13
State listed: Near threatened	5
State listed: Priority 4: Rare Near Threatened and other species in need of monitoring	11
State listed: Other specially protected fauna	1
Least concern	75
Introduced	14

Species labels were selected to balance common, least concern species with less abundant threatened species. As the model does not know what proportion of all species it has been trained on, it is more likely to find familiar features in unknown species rather than leave an animal unclassified. For a rare species label to be useful, the model must also reliably distinguish them from more common species.

Some labels are expected to function as proxies. For example, it was difficult to acquire accurately labelled images of most native rodents due to their similar features and overlapping ranges. It is likely that labels such as 'house mouse' and 'delicate mouse' will act as a proxy for a wide range of native rodent species.



Most images used in model training were sourced from AWC's internal repository of over 50 million images. Additional images were shared by project partners, which yielded a total of 33 unique source locations across Australia. We extracted ~1.7 million images from these datasets. A candidate set of 1.5 million images were verified for correct ID and appropriate quality, resulting in a training set of ~1.1 million images.

Model AWC135 achieved a macro average F1 score of 95% on the top 108 species, and 91% across all 135 species, on the held-out test set. In layman's terms, AWC135 predicted a correct label on average 91% of the time on the verification image set. The model also supports taxonomic fallback, predicting higher-level taxa (e.g., genus or family) when species-level certainty is low (this feature is supported by AddaxAI).

Technical overview

To accurately evaluate the model's real-world performance, we reserved a test set of 76,000 images, representing 7% of the total data. To ensure the model is truly learning to identify species rather than just memorising specific images, our evaluation strategy utilised a temporal split: the test set consisted of images captured at later timestamps and during different surveys than the training data. While these test images were drawn from the same general locations—meaning some degree of background similarity is expected—this approach provided a rigorous evaluation of how the model performs on new events. The performance of each label is available in Appendix 2.

Processing motion-sensor camera data requires filtering out massive amounts of visual noise. These cameras capture everything that triggers their sensors, including empty frames, passing vehicles, or humans. To handle this efficiently, our pipeline is split into two distinct models: an initial detector to find the subject, and a secondary classifier to identify the species.

Stage 1: Animal Detection (MegaDetector)

Before we can classify a species, we must first locate it within the frame. We utilize MegaDetector, a comprehensive, open-source model developed by Dan Morris, specifically designed to make conservation efforts more efficient. Built on the YOLO (You Only Look Once) architecture, MegaDetector acts as our initial filter.

As an image passes through MegaDetector's neural network, it learns to recognize increasingly complex visual features—starting with simple edges and textures, and building up to shapes like legs, ears, and fur. Trained on millions of global camera images, its job is to identify three specific classes: animals, humans, and vehicles. The output is a set of bounding box coordinates that pinpoint exactly where the subject is, along with a confidence score indicating the model's certainty. We then crop the image using these coordinates, removing irrelevant background foliage and landscape before passing the isolated subject to the next stage.

Stage 2: Species Classification (AWC135)

The cropped image is then fed into our custom-developed Australian Wildlife Conservancy model, AWC135. This model is built on the EfficientNet-B5 architecture. We utilised a technique called transfer learning: instead of training the model from scratch, we started with a foundation pre-trained on millions of general, everyday images (ImageNet), and then fine-tuned it specifically on our 1.1 million wildlife images. As the cropped image moves through the model's convolution and pooling layers, feature extraction occurs. Early layers

pick up basic patterns, while deeper layers learn species-specific traits like unique ear shapes, body sizes, or distinct fur patterns.

The model ultimately outputs a probability score for each of the 135 species; essentially it is calculating: "Based on these visual features, how likely is this to be species X?" Typically, we use the top prediction as the species ID, and apply a confidence threshold which allows us to flag uncertain or difficult classifications for manual review.

Model Selection and Optimisation

Choosing the right architecture requires balancing accuracy with computational efficiency. During development, we evaluated several highly complex models, including EfficientNet-XL, Vision Transformers (ViT), and ConvNeXt. By plotting the number of model parameters against the F1-macro score, we observed a clear point of diminishing returns. We selected EfficientNet-B5 because its 30.4 million parameters provide the ideal balance—it is large enough to capture the fine, complex details required to differentiate 135 labels, but efficient enough to deploy and run at scale without unnecessary computational overhead.

Workflow recommendations

Overview

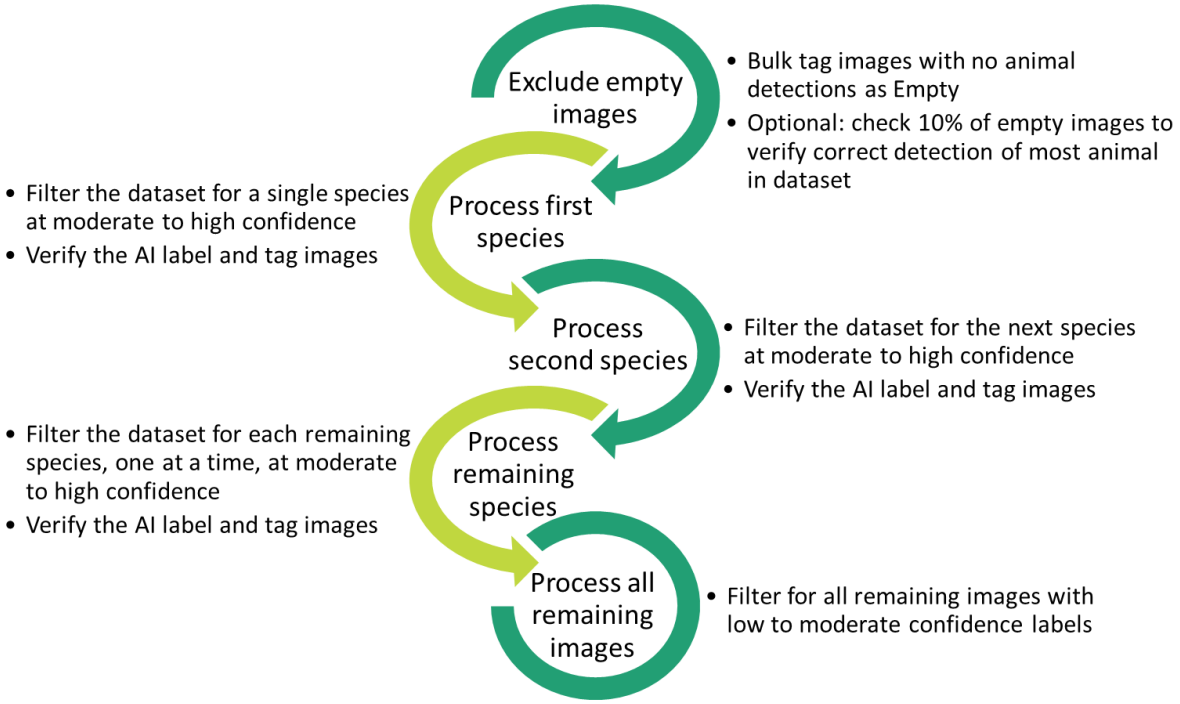
Classification results can be used to increase processing efficiency in a number of ways. The first thing to consider is that AI models are not perfect (see “Limitations” above). Therefore, some level of verification of the labels is essential. The next thing to consider is how comfortable you are with learning to use new programs; Timelapse is the program we recommend for processing motion-sensor camera images because it has many functions that allow for efficient labelling.

The Timelapse function that will increase efficiency the most, is the highly customisable “Custom selection” menu, which allows for filtering and viewing of a subset of images from the full dataset. Images can be selected based on their detection result, their species classification result, or an endless number of other options based on the image metadata and your processing input.

The simplest method is to filter the dataset for images that the MegaDetector has identified as containing an animal, and labelling the images from start to finish in one bulk set. This excludes the empty images, and the species label is still visible for each detection to assist with identification. This approach may be a useful for volunteers or for small datasets. Whilst this method requires the fewest steps and least Timelapse knowledge, it is also the least effective way to utilise the results.

Individual image focus

The classification label is simply a tool to aggregate visually similar animals and images. To get the most out of the labels, it is best to rethink the usual flow of viewing and labelling from start to finish, and instead think about breaking the image set down into visually similar groups, and then filtering and labelling each portion of the images one at a time. We can build on from the first method, excluding empty images by filtering for no detection and marking these as Empty, and apply the same filtering concept to the classification results. The workflow follows the diagram below, with each classification label used to filter a portion of the total image set.



What this looks like in practical use is: filter the dataset for images of Red Kangaroos. Scan through the images and verify that they contain Red Kangaroos, making corrections where needed. Then filter the dataset for Grey Kangaroos; viewing, labelling, and correcting as needed. After that, the dataset could be filtered to select images with a Corvid species label. One by one, the most abundant species can be filtered and processed. Once the bulk of the images are labelled, it may become more efficient to simply select all remaining images in one final step.

To filter the images with a species label, a confidence range must be used to define the selection query. There is no right or wrong range to use, simply a personal preference for how you wish to process your images. One method would be to use a high confidence range (i.e. 0.80–1.00) for each label, and then view all remaining images in a large batch at the end. Alternatively, some people prefer to use multiple confidence range selections for each label, working down in confidence before moving on to the next species label.

This would look like selecting all images with a ‘red kangaroo’ label at very high confidence value (0.95–1.00), verifying them, then selecting all images with a ‘red kangaroo’ label with high confidence values (0.80–0.95), verifying and labelling, then selecting all images with a ‘red kangaroo’ label with a moderate confidence value (0.50-0.80), verifying and labelling, and possibly also selecting low confidence ‘red kangaroo’ labels (0.30-0.50). And finally, some people prefer to select all images with a particular species label at once, regardless of the confidence value. As classifications below 0.30 are removed from the recognition results, this would look like a selection for all images with a ‘red kangaroo’ label (0.30–1.00).

As mentioned earlier, the confidence value is associated with how sure the AI model is in that classification label. High confidence labels are mostly correct, and as you go down in confidence, more incorrect classifications will appear in the images. What this looks like when you select images using one species label, is that the lower you go in confidence value, the higher the number of other animals being included in your filter selection through these misclassifications. Each incorrect label is a small disruption to the flow of verifying the species identification. This is why some people prefer to select multiple confidence ranges per species label, as the high confidence images can be verified very quickly, but the lower confidence images may require slowing down.

The important thing to consider here is that all images are treated as individual entities and are being filtered in isolation to trigger sequences. Each custom selection is returning only those individual images that meet the selection criteria. Therefore, the more restrictive the confidence range used, the more likely it is that images from a trigger event will be separated across multiple selections. And any incorrectly labelled images from a trigger sequence will be shown in the selection for that other label entirely. Keep this in mind if it is important to ensure the entire sequence is viewed for each trigger event.

Using an individual image approach to the selection workflow has several advantages. You will view the smallest number of empty images with this method, as you are only selecting and viewing images with a reasonable detection confidence and classification label. Use of a wide confidence range can help compare visually similar species that may be incorrectly labelled as each other. Sometimes small differences in species are clearer when images are viewed side by side. This approach also works for larger or more easily identifiable species, as they are more likely to be captured in the image in a way that allows AWC135 to confidently and accurately label them.

If your goal is to get general activity data from the cameras and fine grain errors are permissible, then the individual image approach is a good way to verify and label images as quickly as possible.

Event/episode focus

There are circumstances where it is important to view the whole trigger sequence of images. Many species can be difficult to identify from one image alone, particularly when they are small, fast, or too close to the camera for a full body image. In these cases, multiple images from the same trigger event may be required to confidently confirm the species identification.

AI models cannot incorporate this sequence information into their predictions. Each image is classified independently, and labels are assigned solely on the visual information available in each individual frame. However, Timelapse has a function that can group images together based on a customisable time threshold. Images are considered part of the same 'Episode' when the time between them is less than the selected threshold. A new episode begins when the time between two images is more than the threshold value. For example, with a one minute threshold, any two images that are 59 seconds apart (or less) will be considered related and placed in the same episode. Once there is a break of one minute or more, the next image will be considered the start of the next episode. Definitions of 'event' or 'episode' vary in the literature on motion-sensor cameras. Episode groupings can be used either for analytical purposes or simply to assist with image verification and review.

An episode threshold of 2–3 seconds is generally effective for separating images into their trigger sequences. If a camera is immediately triggered again, multiple trigger sequences will be grouped in the same episode. A shorter threshold does run the risk of images within a trigger sequence being separated, as Timelapse rounds milliseconds to the nearest second, and cameras will occasionally have a brief delay between images. Remember: the episode threshold is the time between two images that must be met for a new episode to be created.

So what does this data actually let us do? Using episode data changes how selections are applied. Although labels are predicted independently for each image, Timelapse can be configured to return entire episodes when at least one image within the episode meets the selected criteria. For example, instead of displaying only images with a high confidence classification, every episode that contains at least one high confidence label for that species is displayed.

The general workflow is similar to the individual image focused method. Each label is still verified as chunks of the total dataset. However, episode-based filtering will return more images per selection because each qualifying image brings in its associated images. For this reason, it is recommended to stick to higher confidence ranges to minimise the number of incorrect classifications within each selection.

For example, selecting a very high confidence range (0.95–1.00) for the 'quoll sp' label, will return every episode that has at least one image that contains a 'quoll sp' label between 0.95 and 1.00 confidence. Within each of the episodes, there will be at least one image with

a very high confidence label, likely one or two images with a moderate to high confidence 'quoll sp' label, and potentially an empty image or an image of a Quoll that has an incorrect label.

By contrast, selecting a moderate confidence range (0.50–0.80) increases the likelihood that an episode is selected due to a single incorrect prediction. This may result in entire episodes of another species being included because one image was misclassified at moderate confidence. For example, if a handful of cat images were incorrectly classified with the 'quoll sp' label at 0.55 confidence, the whole episode for each would be included in the selection for 'quoll sp' labels between 0.50 and 1.00 confidence. A practical approach is therefore to verify high-confidence predictions for each species first (e.g. high-confidence 'quoll sp', then high-confidence 'domestic cat'), and only then move to lower confidence ranges.

Episode-based filtering can help smooth out single incorrect labels, as a single misclassification may be reviewed alongside correctly labelled images in the same episode. However, if confidence thresholds are too broad, more incorrect species may be included in the returned selections.

It is generally advisable to start with the most abundant species, or label that is most likely to be correct (this will vary by dataset). Processing these first allows any incorrect labels within the episodes to be processed and set aside, reducing the likelihood that they will affect subsequent filtering steps for other species.

Selecting whole episodes will include empty images where the animal has moved out of frame, adding to the total number of images viewed. However, this approach has the advantage that any images of animals that the MegaDetector missed will still be displayed, as long as at least one image in the episode contains a detected bounding box.

This method is particularly useful when it is important that every image of an animal is viewed, as the majority of missed animal detections are within trigger sequences that include at least one correctly detected animal bounding box. It is also well suited to workflows where only a single species record is required per trigger or episode, as all related images are viewed together. Finally, if individual animal identification is possible, episode-based filtering keeps images of the same animal together in the same selection.

Workflow example

The two suggested workflow methods can be used together, and there are as many slight variations as there are people using Timelapse. There is no one correct way to use the AI outputs, provided you understand what each filter step is expected to return, and what may be missed or incorrectly included.

Regardless of how you choose to filter your images, it is important to have a field in Timelapse that you can use to keep track of images that you have reviewed. The “Species” field can serve this purpose, however, keep in mind that you may not be entering a value for every image, especially for datasets that only require one species label per trigger sequence/episode. “Animal”/“Empty” fields can be useful and intuitive, however, this requires a value to be assigned to each image or adjusted during verification.

An alternative is to use a field such as “Processed by” and enter information as the final component of each filtering step. Because the value will be consistent across the selected batch, it can be entered once and copied to all images in that selection. This field can then be used to exclude images that were already viewed from future filtering selections.

Fields used in Timelapse are customisable through the Template Editor program installed with Timelapse, allowing these options to be trialled and adjusted to fit your own preferences.

A workflow example is given below to help familiarise you with using recognition data. Please refer to the Basic Timelapse Functionality for more information, or head straight to the comprehensive Timelapse user guides.

- Load in images, extract metadata, create episode data
- Run AddaxAI and load in recognition file
- Use the Custom selection menu, select ‘Empty’ and set the confidence range to 0.20–1.00. Confirm selection.
- Tick the “Empty” field, copy to all. Enter name in “Processed by” field, copy to all. (Optional: quickly scan through ~10% of the images to calculate an error rate of missed animals.)
- Use the Custom selection menu, select ‘animal’ and set the confidence range to 0.20–1.00. Use the “Update counts” button to identify the most common classification labels in your dataset. Select the classification label with the highest count, and set the confidence range to 0.95–1.00. Tick “Include all files in an episode when at least one file matches”. Tick next to the field “Processed by” and leave blank. Confirm selection.
- On the first correct image, type/select the species name in the “Species” field. Scan through the images until an empty image or incorrect label is found. Go to the last correct image, and use “Propagate from the last non-empty value to here” on the “Species” field. On the next correct image, type/select the species name into the “Species” field, scan till the next break in correct images, on the last correct image propagate from last species entry. Continue on, bulk entering the correct label in the species field and individually entering any other species names. (Alternatively: enter the species name once, on first or best image per episode.) On the final image, enter name in “Processed by” field, copy to all.

- If entering a species value for every image: Use the Custom selection menu, select the classification label with the next highest count, and set the confidence range to 0.80–1.00. Untick “Include all files in an episode when at least one file matches”. Leave tick next to the field “Processed by” and leave blank. Confirm selection.
- On the first correct image, type/select the species name in the “Species” field. Copy to all. Scan through images and change the information in the species field when necessary. On the final image, enter name in “Processed by” field, copy to all.
- If only entering a species value once per episode: Use the Custom selection menu, select the classification label with the next highest count, and set the confidence range to 0.95–1.00. Leave “Include all…” and blank “Processed by” field ticked. Confirm selection.
- Continue on, alternating between individual image or whole episode selections depending on preference and species labelling method. Leave tick next to the blank “Processed by” field and select ‘animal’ and “Update counts” to check how many total images are left, and how that is spread across the unprocessed classification labels.
- When all relevant labels have been processed at high to very high confidence, go back through each, using a lower confidence range in each selection (e.g. 0.80–1.00 or 0.50–0.80).
- When labels containing the bulk of images have been processed, use the Custom selection menu, select ‘animal’ and set the confidence range to 0.20–1.00. Leave blank “Processed by” field ticked. Confirm selection.
- This will return all images with an animal detection result that have not already been viewed. Classification labels will be visible. Utilise the “Propagate from last…” and Quick Paste functions for this selection. On the final image, enter name in “Processed by” field, copy to all.
- Reset to view all images.
- Use File menu and select “Export image/video data in the current selection to a .csv file…”. Select location and save.
- Highly recommended: Open csv file and save as excel workbook immediately. Store images, Timelapse files, and csv file together in the same folder structure for long term storage.

Simple step-through guide

Getting set up

Download Timelapse

https://timelapse.ucalgary.ca/

Timelapse is the program that is recommended for processing motion-sensor camera images. It can view images and read AI recognition information, and has many functions that allow for efficient labelling. On the website you will find information about downloading the program, and very detailed tutorials. The Quick Start Guide will show you how to load your images for the first time. Please refer to the additional user guides for a full understanding of Timelapse's functionality.

Download AddaxAI

https://addaxdatascience.com/

AddaxAI is how you will access AWC135. This software allows you to run the AI model on your local desktop. On the website you will be able to learn more about its functions and supported models, and download the software. Download is also available through the Timelapse program. AI models are downloaded upon use. As the processing is done locally, it relies on the computing power of the desktop/laptop that you use. Most laptops will process the images quite slowly, so we recommended setting up the AI processing on a desktop computer that has a decent GPU. Refer to the supporting documentation for more information on processing speeds.

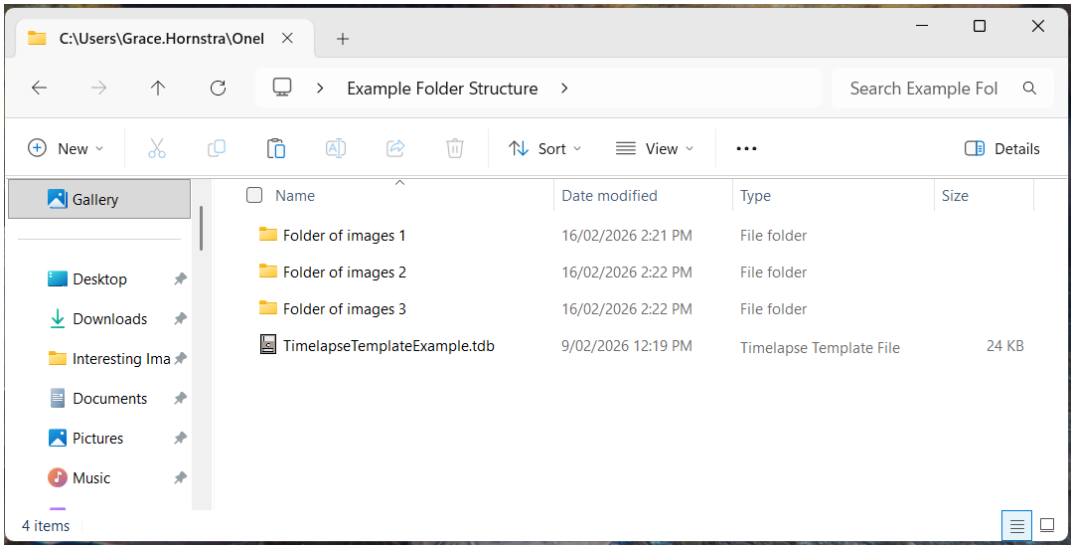
Download the example template file

https://github.com/Australian-Wildlife-Conservancy-AWC/awc-wildlife-classifier/raw/refs/heads/main/timelapse_templates/TimelapseTemplateExample.tdb

A template file (.tdb) is required to load new images into Timelapse. When new images are loaded in, Timelapse will pull out date and time metadata for each image and put it into a data table. The template file is what creates the structure to build the data table, with each editable field equalling a column in the data table. Each field can be added or removed from the template file, allowing for full customisation of data processing. An example template has been provided on the GitHub page for download to get you started. Another example template can be found on the Timelapse webpage.

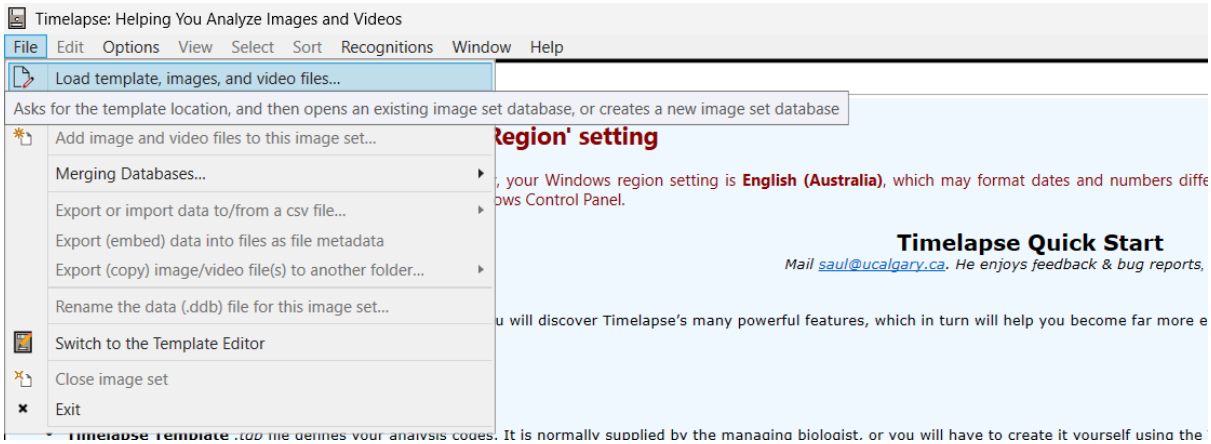
Set up your folder structure

Make sure your folder structure looks like this example. The folder “Example Folder Structure” is called the root folder. Place your template file inside the root folder, alongside your folder/s of images. Images can be stored in either one or multiple folders, or can be stored in multiple subfolders. When images are first loaded in, a Timelapse database file (.ddb) and a “Backups” folder will be created and saved alongside the template file.



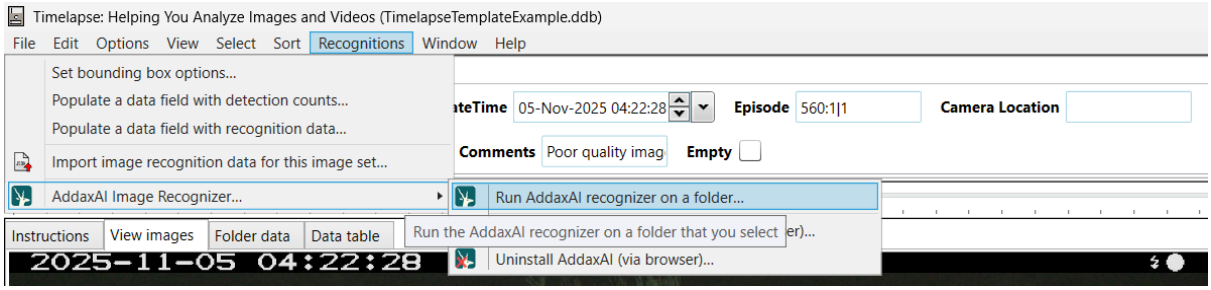
Open Timelapse and load in your images

Use the "Load template, images, and video files…" option and select the template file. Timelapse will automatically load in all image files that are contained in folders (and subfolders) alongside the template file. Follow Timelapse's Quick Start Guide for more information.



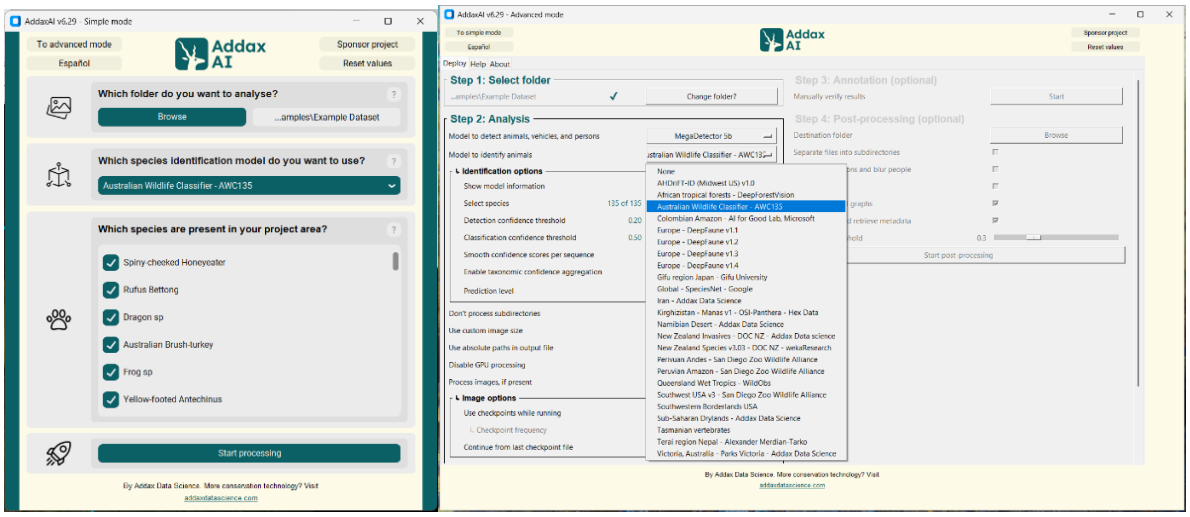
Open AddaxAI

Use the "Recognitions" menu and select "Run AddaxAI recognizer on a folder". Select the folder that you wish to run through the AI (all files in subfolders will be processed). If you have multiple folders sitting next to your Timelapse files, you can go up a level and select the root folder. If you have a very large dataset that is broken up into numerous folders, you may wish to run each through the AI separately. You can load several recognition files into Timelapse.



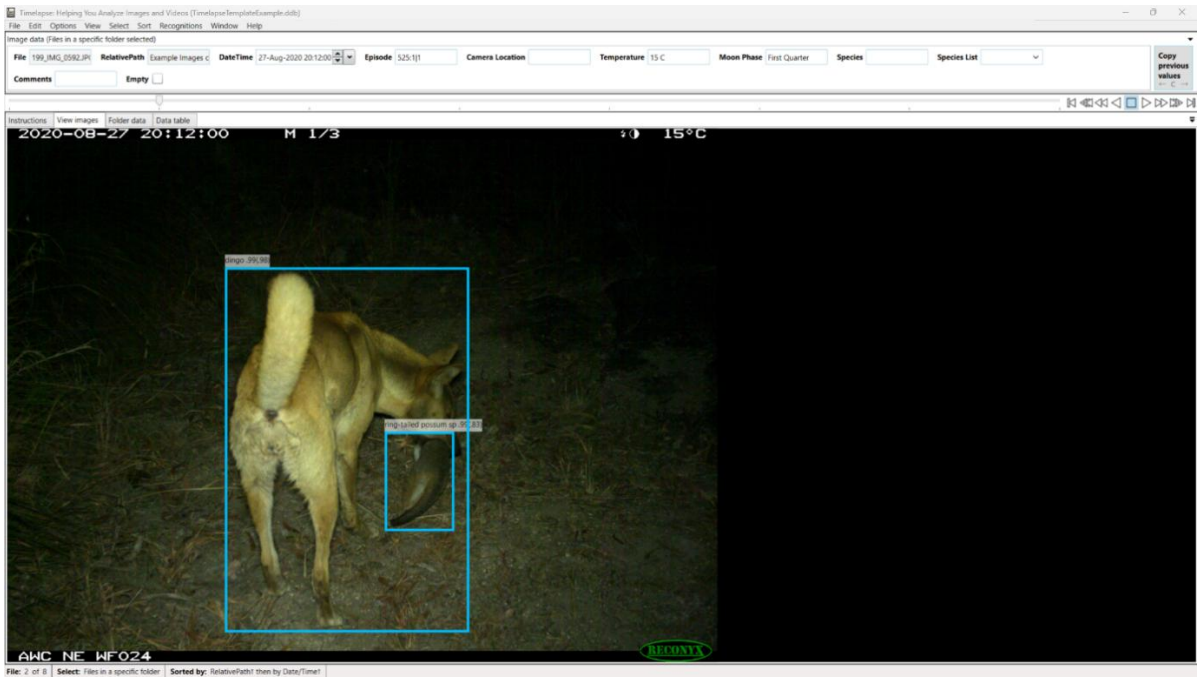
Select Australian Wildlife Classifier - AWC135

The user interface of AddaxAI has a 'Simple' and 'Advanced' option. The 'Advanced Menu' will provide more customisation and usability. Select 'Australian Wildlife Classifier – AWC135" as the classification model. View the "Help" tab for an explanation of all customisable options. When you are satisfied with the selections, scroll to the bottom of the page and press "Start processing".



Load the recognition data

Use the "Recognitions" menu and select "Import image recognition data for the image set" and select the "timelapse_recognition_file.json" file. This file will be saved **inside** the folder that was selected earlier. The loaded recognition information will appear visually over your images.

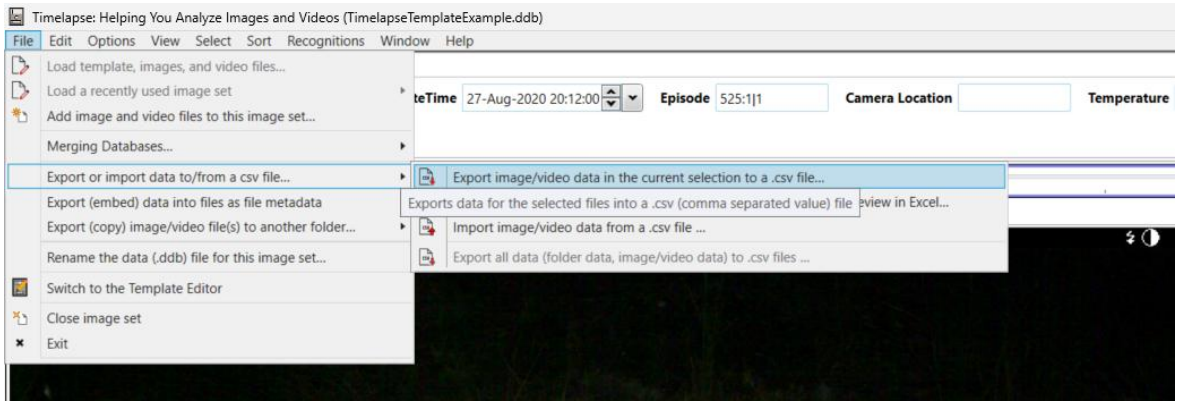


Process those images!

There are as many methods to image processing as there are people doing the work. Watch the webinar or refer to the supplementary documentation for some suggested workflows and tips/tricks for utilising Timelapse's functions for efficient viewing and labelling. Some suggested functions to learn are: populating a field with metadata, utilising the custom selection menu, creating episode data, the QuickPaste window, and the different options for bulk filling fields. Refer to Timelapse's "Reference Guide" for a full description of functions that will assist with speedy processing. Refer to Timelapse's "Image Recognition Guide" for information on using the AI model results and additional processing suggestions.

Export your finished data to a .csv file

Once you are happy that your dataset is finalised and all labelling/data input is complete, use the "Export image/video data in the current selection to a .csv file…" option to generate a spreadsheet of your data. Make sure that you have reset your selection of images back to "All files" if you want the full dataset. Please remember: save your .csv file as an excel workbook when you first open it to ensure you can save multiple tabs!!



Basic Timelapse functionality

https://timelapse.ucalgary.ca/

Timelapse is updated fairly regularly, so please refer to the user guides for the most up to date information. The following functions are recommended to improve processing speed.

Populate fields with metadata – allows for extraction of information from the metadata of each image, such as temperature, moon phase, camera make/model, etc.

Populate a field with episode data – creates episode data and fills into specified field (threshold is adjusted under Options).

Custom selection menu – very powerful menu for filtering the dataset for images that meet the selection criteria. Images can be selected based on recognition information (detection or classification), and/or information in the available fields. Most fields have an option to select = or ≠ whatever input term is chosen. Multiple selection criteria can be combined as an "And" or "Or" condition.

> *Select recognitions* – uses the recognition information to select images from the full dataset. Selected categories are highlighted, confidence range determines criteria. Detection confidence will affect classification category result, however, classification confidence will not affect detection search criteria.
> *Update counts* – shows how many images contain each category's label.
> Include all files in an episode… – select individual images that meet the selection criteria, or the whole episode when at least one image meets the criteria.

Propagate from the last non-empty value to here – quick way to bulk fill the same value in a selected field, between the last entry of data and current image. Used on individual fields at a time, filling with the last value entered up to the current image.

Copy to all – quick way to bulk fill the same value in a selected field to all images in the current selection. If used on a custom selection portion of the dataset, it will only affect those selected images. Blank fields can be copied to all, useful to reset the data but cannot be undone. A popup will ask for confirmation.

Quick Paste – create 'profiles' to fill data into multiple fields when shortcut keys are selected. Can be as simple as just the common names for the most abundant species, or include common name, scientific name, Animal box ticked, etc. The first 10 Quick Pastes created are given shortcut keys; additional can be clicked on within the Quick Paste Window to fill the values into data fields.

FAQs

Question: Is it possible to train the AWC135 model on species that are not currently included? What is the mechanism for ongoing training of the classifier?

Answer: Model AWC135 (and all AI models available through AddaxAI) is a closed system, completed model. To add any additional information requires full training of an entirely new model. For this, all training images used in the current model would need to be available, along with the new images. As our source image library is not publicly available, this cannot be done by external parties. AWC is open to consider partnerships or collaboration activities that result in new, custom, lightweight or expanded models to include additional or custom species. If this is of interest, please email ai.info@australianwildlife.org and provide details.

Question: Does the performance vary by habitat? Does the accuracy decrease in off-track, shrubby habitat where it can be hard for manual users to see 'hidden' wildlife, or only a small part of their body is visible?

Answer: We have attempted to include a wide varity of training images for the species labels in AWC135, including images where only a portion of the animals body is shown. The key factor was that the animal still had to be identifiable at least to its overarching group (i.e. Red Kangaroo images had to still be easily identified as a macropod). The fewer key characteristics of the animal that is visable, the harder it is for the model to predict the label, and the confidence and performance drops compared to full body images. Presume that if it is difficult for an average person to correctly identify an animal, it will also be difficult for the model. If the background features of your dataset look completely different to the training data, this will impact the performance. When the animal is very difficult to see, there is a chance that the MegaDetector will miss them, and therefore model AWC135 will not be able to attempt to label them.

Question: Is it possible to implement this on programs other than Timelapse, and if so, what would be the requirements of a program to do this?

Answer: Timelapse is used to visualise the result of the detection/classification model, and it's a very efficient tool to visualise large amounts of images/detections in a short amount of time, so we have not looked at alternatives. For the model itself (AWC135), it is possible to run it in Python, which requires a bit of coding and some package installation from the GitHub page. Our second webinar covered the use and implementation of running the model this way.

Question: Is the limiting factor for the number of species (labels) that can be identified by the model due to the resources needed to train the model (availability of training images, verification effort), or does increasing the number of labels have unintended consequences for model accuracy/confidence?

Answer: Increasing the number of labels should not affect the model accuracy too much, since modern computer vision architectures have the capacity to predict +2000 classes. However, if the training data is not high enough quality, the model may be forced to include non-animal information into the parameters for classification. So while the AI is capable of predicting many more classes, the difference between those classes/labels becomes the limiting factor for model accuracy. For example, there are many species that are incredibly difficult to identify in camera images, or species that are distinguished by geographical location rather than visual phenotypes. The inclusion of many of these species will have an impact on the model's performance.

Question: What computer specifications do you recommend for using AddaxAI?

Answer: Any standard laptop will be powerful enough to run Timelapse, however, runnning AI models takes a decent amount of computing power. Laptops or desktops with a graphics card will process the images faster than using the CPU. For datasets with tens of thousands images, a desktop computer with a decent GPU is recommended. A list of processing speeds across several example devices has been listed in the appendix.

Question: It seems that it would be fairly easy to add a species distribution sort of filter to the data, based on location of the image. Is that an upgrade that is being considered?

Answer: Currently, the only image recognition architecture available to build upon is this 'simplistic' process of utilising purely visual information from the images to create rules and parameters for prediction on new images. Incorporating geographic information into the model prediction would essentially mean creating a brand new kind of model, with no building blocks or tools to get started. There is the capacity to unselect species labels before running the model in AddaxAI, however, this is a basic logic function that removes any labels that have been unselected. The model prediction is not influenced by this, and the removed label is not replaced with a "more likely" species label. These labels have no classification information and are left as "animal" detections, which means that they cannot be filtered into a group of similar images for speedy verification. Incorrect labels are still useful as they do allow for this grouping of similar images, and predicted label information does not have to be the species information that is collected.

Question: Can you 'upgrade' the species label that you then export into your csv files? I.e. you have 'Quoll sp' but can you then label it 'Western Quoll' after reviewing, if you know that is the species at that location? Or e.g. 'Bandicoot' labelled as 'Quenda'?

Answer: Remember that the label that the AI produces is simply a tool for aggregating your images for faster verification and data entry. If you wish to use the label output without verification, you can always change the species label in your spreadsheet. However, it is not recommended that just trust the output 100%, as there are numerous ways mistakes can creep into the data. No matter what label the model predicts for the images, the data entry should be done by a person, and you can therefore input any species name you want.

Question: Has this model been tested against other models and how well did it performed?

Answer: There are currently no other Australia-wide classification models to test AWC135 against. There are a few habitat/location-specific models which are likely to outperform AWC135 in those specific circumstances, but are not applicable in other regions. SpeciesNet is a coarse-grain 'global' classifier, however, the training data for Australian species is under represented and the model does not perform particularly well on our animals.

Question: Have there been any trials testing this on camera trap video data? If so, how did that go?

AddaxAI has the capacity to run videos through the AI models in a similar way to images. It does this by taking a still frame every few seconds (this is an adjustable setting), and using that still frame as an image to be classified. No testing has been done on this function specifically, however, the AI still operates the same so a similar performance can be expected.

Appendix

Appendix 1: Links to resources

Timelapse:
https://timelapse.ucalgary.ca/

AddaxAI:
https://addaxdatascience.com/

Australia Wildlife Classifier – AWC135 (GitHub page):
https://github.com/Australian-Wildlife-Conservancy-AWC/awc-wildlife-classifier

Example Timelapse template file for download:
https://github.com/Australian-Wildlife-Conservancy-AWC/awc-wildlife-classifier/raw/refs/heads/main/timelapse_templates/TimelapseTemplateExample.tdb

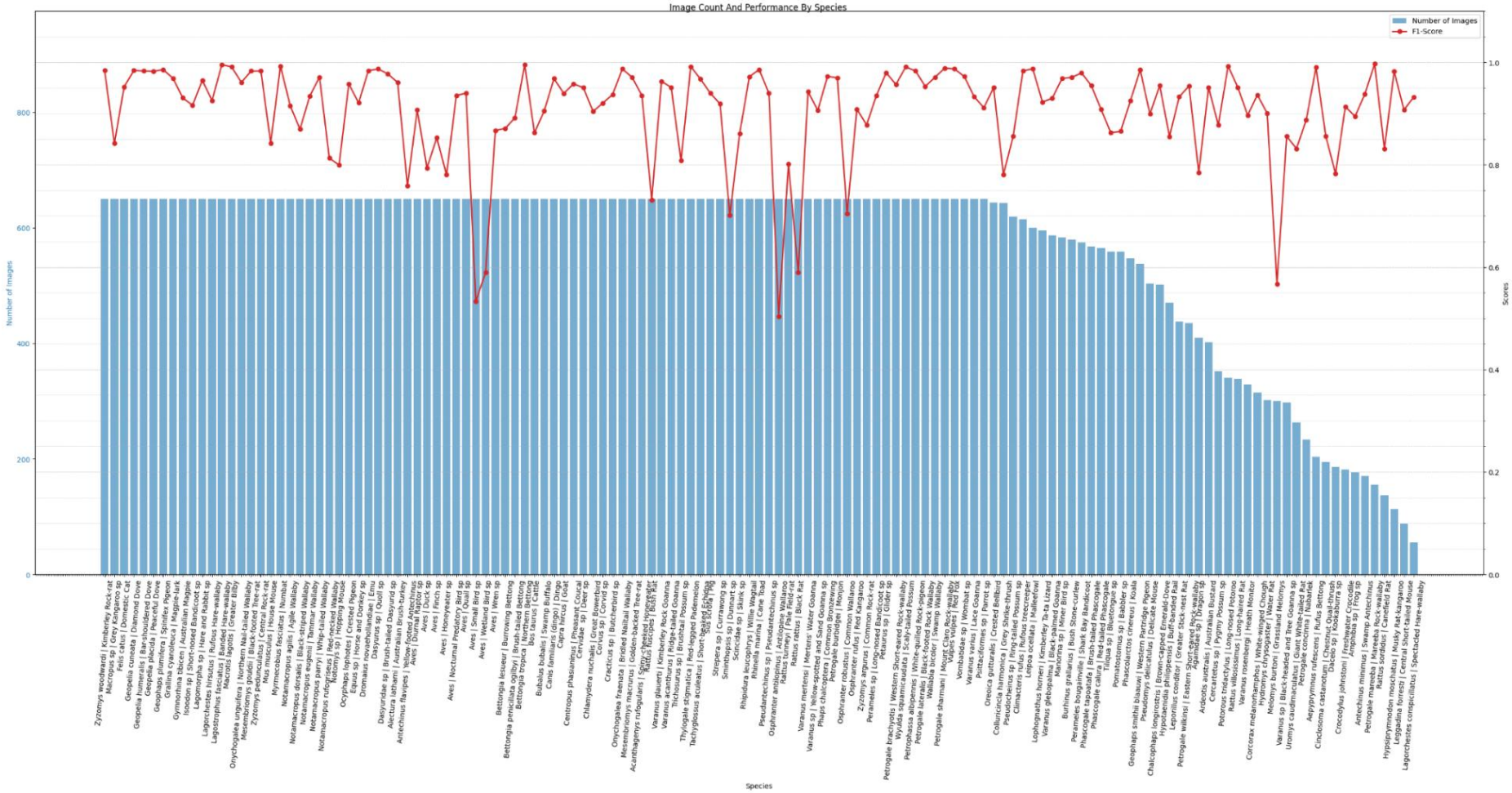
MegaDetector User Guide:
https://github.com/agentmorris/MegaDetector/blob/main/megadetector.md

Blog post explaining MegaDetector versions:
https://github.com/agentmorris/MegaDetector/blob/main/docs/release-notes/mdv1000-release.md

AddaxAI forum
https://forum.addaxai.com/c/general/4

Appendix 2: Model performance graph

The performance of each label was tested with a verification image set containing a stratified sample of images per label (blue bars), and the proportion of correct predictions of labels was calculated using an F1 score (red line). Verification images were sourced from the candidate dataset and were not used in model training.



Appendix 3: Full list of labels

Introduced
Black Rat
Cane Toad
Cattle
Deer sp
Domestic Cat
Goat
Hare and Rabbit sp
Horse and Donkey sp
House Mouse
Pig
Red Fox
Swamp Buffalo

Mammal
Agile Wallaby
Antilopine Wallaroo
Banded Hare-wallaby
Black-footed Rock-wallaby
Black-footed Tree-rat
Black-striped Wallaby
Bridled Nailtail Wallaby
Brushtail Possum sp
Brush-tailed Bettong
Brush-tailed Dasyurid sp
Brush-tailed Phascogale
Burrowing Bettong
Bush Rat
Canefield Rat
Central Rock-rat
Central Short-tailed Mouse
Common Dunnart
Common Rock-rat
Common Wallaroo
Delicate Mouse
Dingo
Eastern Short-eared Rock-wallaby
Giant White-tailed Rat
Glider sp
Golden-backed Tree-rat
Grassland Melomys
Greater Bilby
Greater Stick-nest Rat
Grey Kangaroo sp
Hopping Mouse
Kimberley Rock-rat
Koala
Long-haired Rat
Long-nosed Bandicoot
Long-nosed Potoroo
Mareeba Rock-wallaby
Monjon
Mount Claro Rock-wallaby
Musky Rat-kangaroo
Nabarlek
Northern Bettong
Northern Nail-tailed Wallaby
Numbat
Pale Field-rat
Pseudantechinus sp
Pygmy Possum sp
Quoll sp
Red Kangaroo
Red-legged Pademelon
Red-necked Wallaby
Red-tailed Phascogale
Ring-tailed Possum sp
Rufous Hare-wallaby
Rufus Bettong
Scaly-tailed Possum
Shark Bay Bandicoot
Short-beaked Echidna
Short-nosed Bandicoot sp
Spectacled Hare-wallaby
Swamp Antechinus
Swamp Wallaby
Tammar Wallaby
Water Rat
Western Short-eared Rock-wallaby
Whip-tailed Wallaby
Wombat sp
Yellow-footed Antechinus

Reptile/Amphibian
Black-headed and Tree Goanna sp
Black-palmed Goanna
Bluetongue sp
Dragon sp
Freshwater Crocodile
Frog sp
Heath Monitor
Kimberley Rock Goanna
Kimberley Ta-ta Lizard
Lace Monitor
Mertens' Water Goanna
Ridge-tailed Goanna
Skink sp
Yellow-spotted and Sand Goanna sp

Bird
Australian Brush-turkey
Australian Bustard
Australian Magpie
Babbler sp
Bar-shouldered Dove
Brown-capped Emerald-Dove
Buff-banded Rail
Bush Stone-curlew
Butcherbird sp
Chestnut Quail-thrush
Common Bronzewing
Corvid
Crested Bellbird
Crested Pigeon
Currawong sp
Diamond Dove
Diurnal Raptor sp
Duck sp
Emu
Finch sp
Great Bowerbird
Grey Shrike-thrush
Honeyeater sp
Kookaburra sp
Magpie-lark
Malleefowl
Miner Bird sp
Nocturnal Predatory Bird sp
Parrot sp
Peaceful Dove
Pheasant Coucal
Quail sp
Rufous treecreeper
Small Bird sp
Spinifex Pigeon
Spiny-cheeked Honeyeater
Western Partridge Pigeon
Wetland Bird sp
White-quilled Rock-pigeon
White-winged Chough
Willie Wagtail
Wren sp

***Bold headings are not labels**

Appendix 4: Timing results for AI processing

Timing results for MDv5 and MDv1000-redwood only (read more at: https://github.com/agentmorris/MegaDetector/blob/main/megadetector.md#how-fast-is-megadetector-and-can-i-run-it-on-my-giantsmall-computer)

- An RTX 4090 processes around 17.6 images per second, or around 1,500,000 images per day (for MDv5/MDv1000-redwood)
- An RTX 3090 processes around 11.4 images per second, or around 985,000 images per day (for MDv5/MDv1000-redwood)
- An RTX 3080 processes around 9.5 images per second, or around 820,800 images per day (for MDv5/MDv1000-redwood)
- A desktop RTX 3050 processes around 4.2 images per second, or around 363,000 images per day (for MDv5/MDv1000-redwood)
- A laptop RTX 3050 processes around 3.0 images per second, or around 250,000 images per day (for MDv5/MDv1000-redwood)
- A Quadro P2000 processes around 2.1 images per second, or around 180,000 images per day (for MDv5/MDv1000-redwood)
- A 2024 M3 MacBook Pro (18 GPU cores) averages around 4.61 images per second, or around 398,000 images per day (for MDv5/MDv1000-redwood)
- A 2020 M1 MacBook Pro (8 GPU cores) averages around 1.85 images per second, or around 160,000 images per day (for MDv5/MDv1000-redwood)
- An Intel Core i7-12700 CPU processes around 0.5 images per second on a single core (43,000 images per day) (multi-core performance is... complicated) (for MDv5/MDv1000-redwood)
- An Intel Core i7-13700K CPU processes around 0.8 images per second on a single core (69,000 images per day) (multi-core performance is... complicated) (for MDv5/MDv1000-redwood)

Timing results for running both MegaDetector and AWC135, using scripting command line. Please note: running the AI through AddaxAI will be ~20% slower.

Device	Throughput (images/second)	Time to process 1 million images
CPU (i7 9750h, single core, standard in most work laptops)	0.29 (or 3.4 seconds per image)	40 days
GTX 1650 (standard in high powered/gaming laptops)	2.46	4.7 days
RTX 3050 (mid-range GPU for desktop computers)	3.97	2.9 days
RTX 5070 (high-end GPU for desktop computers)	8.39	1.38 days (33 hours)
A100 (offsite GPU for cloud processing)	28	0.41 (9.9 hours)
H100 (offsite GPU for cloud processing)	34	0.34 (8.1 hours)

Appendix 5: Description of training data for each label

Species Label	% of colour images	% of black and white images	Description of camera set up	Description of background context
Agile Wallaby	98.8	1.2	Mix of front facing (perpendicular and angled orientation)	Leaf litter, grass, bare soil
Antilopine Wallaroo	99.0	1.0	Mostly front facing, angled orientation	Grass, bare soil, rocky slab
Australian Brush-turkey	99.8	0.2	Mix of front facing (perpendicular and angled orientation)	Brown leaf litter, some shrubbery
Australian Bustard	97.8	2.3	Mix of front facing (perpendicular and angled orientation)	Bare soil, dry grass, some trees and sky
Australian Magpie	99.3	0.7	Mix of front facing (perpendicular and angled orientation)	Grass, leaf litter, bare soil, some images contain green water trough
Babbler sp	99.9	0.1	Mix of front facing (perpendicular and angled orientation)	Leaf litter, grass, bare soil, perching stick
Banded Hare-wallaby	1.2	98.9	Mix of front facing (perpendicular) and downward facing	Bare soil, grass, wooden checkered board
Bar-shouldered Dove	100.0		Mostly front facing, angled orientation	Leaf litter, grass, rocks
Black Rat	31.4	68.6	Mostly front facing, perpendicular to ground	Grass, bare soil, shrubbery, bait canister and pole occasionally present
Black-footed Rock Wallaby	14.2	85.8	Mostly front facing, perpendicular to ground	Rocky slab, rocks, grass
Black-footed Tree-rat	96.5	3.5	Mostly front facing, angled orientation	Bare soil, grass, rocks, bait canister and pole occasionally present
Black-headed and Tree Goanna sp	94.8	5.2	Mix of front facing (angled orientation) and downward facing	Leaf litter, tree trunk, rocks, bare soil
Black-palmed Goanna	98.7	1.3	Mix of front facing (angled orientation) and downward facing	Rocky slab, rocks, leaf litter
Black-striped Wallaby	88.0	12.0	Front facing, perpendicular to ground	Grass, bare soil, leaf litter, trees
Bluetongue sp	96.1	3.9	Mix of front facing (perpendicular and angled orientation) and downward facing	Leaf litter, wooden checkered board, rocks, bare soil, grass
Bridled Nailtail Wallaby	80.5	19.5	Front facing, perpendicular to ground	Bare soil, grass, shrubbery, some images contain green water trough
Brown-capped Emerald-Dove	100.0		Mix of front facing (angled orientation) and downward facing	Leaf litter, rocky slab
Brushtail Possum sp	68.3	31.7	Mix of front facing (perpendicular and angled orientation) and downward facing	Leaf litter, bare soil, grass, wooden checkered board, bait canister occasionally present
Brush-tailed Bettong	43.5	56.6	Mix of front facing (perpendicular) and downward facing	Bare soil, some shrubbery, wooden checkered board

Brush-tailed Dasyurid sp	12.3	87.7	Mostly front facing, perpendicular to ground, 5-20cm off ground, some downward facing	bare soil, wooden checkered board
Brush-tailed Phascogale	97.8	2.2	Mix of front facing (angled orientation) and downward facing	Leaf litter, tree trunk, wooden log
Buff-banded Rail	100.0		Mix of front facing (angled orientation) and downward facing	Leaf litter, rocks, bare soil
Burrowing Bettong	35.3	64.7	Mostly front facing, perpendicular to ground	Bare soil, some shrubbery
Bush Rat	42.7	57.3	Mostly front facing, perpendicular to ground	Grass, leaf litter, shrubbery, bait canister and pole occasionally present
Bush Stone-curlew	90.9	9.1	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, grass
Butcherbird sp	98.2	1.8	Mostly front facing, perpendicular to ground, some angled orientation	Leaf litter, bare soil, grass, rocky slab, some images contain green water trough and/or perching stick
Cane Toad	91.8	8.2	Mostly front facing, angled orientation	Bare soil, rocky slab, grass
Canefield Rat	100.0		Mostly front facing, angled orientation	Leaf litter, grass, bait canister occasionally present
Cattle	98.9	1.1	Mix of front facing (perpendicular and angled orientation)	Bare soil, grass (dead and live), trees, sky
Central Rock-rat	54.1	45.9	Mix of front facing (angled orientation) and downward facing	Rocky slab, rocks, grass, bait canister occasionally present
Central Short-tailed Mouse	100.0		Mostly front facing, perpendicular to ground, 5-20cm off ground	Bare soil, some small rocks
Chestnut Quail-thrush	99.0	1.0	Mostly front facing, perpendicular to ground	Bare soil (mostly red), leaf litter, grass
Common Bronzewing	85.9	14.1	Mostly front facing, perpendicular to ground	Bare soil (mostly red), grass, leaf litter, some images contain green water trough
Common Rock-rat	99.2	0.9	Mostly front facing, angled orientation	Rocky slab, rocks, leaf litter, bait canister occasionally present
Common Wallaroo	87.2	12.8	Mix of front facing (perpendicular and angled orientation)	Bare soil, rocks, grass
Corvid sp	98.6	1.4	Mix of front facing (perpendicular and angled orientation) and downward facing	Leaf litter, grass, bare soil, shrubbery, some images contain green water trough or bait canister
Crested Bellbird	96.0	4.0	Mostly front facing, perpendicular to ground	Bare soil (mostly red), grass, leaf litter
Crested Pigeon	99.9	0.1	Mostly front facing, perpendicular to ground	Bare soil (mostly red), grass
Currawong sp	98.8	1.2	Mostly front facing, perpendicular to ground	Grass, leaf litter, bare soil, some images contain green water trough
Deer sp	77.7	22.3	Mix of front facing (perpendicular and angled orientation)	Leaf litter, grass, white food trough occasionally present
Delicate Mouse	99.3	0.7	Mostly front facing, angled orientation	Bare soil, wooden log, grass

Diamond Dove	99.7	0.3	Mix of front facing (perpendicular and angled orientation)	Bare soil (mostly red), leaf litter, grass, rocks
Dingo	91.8	8.2	Mix of front facing (perpendicular and angled orientation), some downward facing	Bare soil, leaf litter, grass, trees, sky, rocky slab
Diurnal Raptor sp	97.8	2.2	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil, dry grass, some sky, star picket occasionally present
Domestic Cat	82.3	17.7	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil, leaf litter, grass, rocks/rocky slab, bait canister and pole occasionally present
Dragon sp	100.0		Mix of front facing (perpendicular and angled orientation) and downward facing	Mixed background, more red/yellow soil than grass
Duck sp	99.3	0.7	Mix of front facing (perpendicular and angled orientation) and downward facing	Leaf litter, bare soil, some water
Dunnart sp	56.0	44.0	Mostly front facing, perpendicular to ground	Grass, bare soil, mesh drift fence
Eastern Short-eared Rock-wallaby	100.0		Mostly front facing, perpendicular to ground	Rocky slab, rocks, bare soil
Emu	99.1	0.9	Mostly front facing, perpendicular to ground, some angled orientation	Grass, sky, trees, bare soil
Finch sp	97.8	2.2	Mix of front facing (angled orientation) and downward facing	Bare soil (brown and red), leaf litter, some grass
Freshwater Crocodile	99.4	0.6	Mix of front facing (angled orientation) and downward facing	Bare soil, rocks, leaf litter, some water, bait canister and star picket occasionally present
Frog sp	93.8	6.2	Mix of front facing (angled orientation) and downward facing	Bare soil, rocky slab
Giant White-tailed Rat	100.0		Mostly front facing, angled orientation	Rocky slab, rocks, leaf litter
Glider sp	100.0		Mostly front facing, angled up tree trunks	Tree trunk, night sky, bark, leaf litter, bait canister occasionally present
Goat	90.1	9.9	Mostly front facing, perpendicular to ground, some angled orientation	Grass, trees, dry ground, shrubbery
Golden-backed Tree-rat	98.1	2.0	Mostly front facing, angled orientation	Rocky slab, rocks, bare soil, leaf litter
Grassland Melomys	100.0		Mostly front facing, angled orientation, some downward facing	Leaf litter, rocks, bare soil
Great Bowerbird	100.0		Mix of front facing (angled orientation) and downward facing	Bare soil, rocky slab, leaf litter, handful of Quail-thrush errors
Greater Bilby	62.7	37.3	Mix of front facing (perpendicular) and downward facing	Bare soil, grass, shrubbery, wooden checkered board, some images contain green water trough or bait canister

Greater Stick-nest Rat	30.7	69.3	Mix of front facing (perpendicular) and downward facing	Bare soil, grass, shrubbery, wooden checkered board, some images contain green water trough or metal cage trap
Grey Kangaroo sp	81.6	18.4	Mostly front facing, perpendicular to ground, some angled orientation	Grass, leaf litter, bare soil, some images contain green water trough
Grey Shrike-thrush	99.5	0.5	Mostly front facing, perpendicular to ground, some angled orientation	Bare soil, leaf litter, grass
Hare and Rabbit sp	54.6	45.4	Mostly front facing, perpendicular to ground, some downward facing	Grass, shrubbery, bare soil, wooden checkered board, bait canister and pole occasionally present
Heath Monitor	100.0		Mostly front facing, perpendicular to ground, 5-20cm off ground	Leaf litter, grass, mesh drift fence, shrubbery
Honeyeater sp	97.7	2.3	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil (mostly red), leaf litter, some images containing green water trough
Hopping Mouse	26.4	73.6	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil, wooden checkered board, leaf litter
Horse and Donkey sp	99.2	0.8	Mostly front facing, perpendicular to ground, some angled orientation	Bare soil, grass, trees, sky, rocks
House Mouse	44.2	55.8	Mostly front facing, perpendicular to ground, 5-20cm off ground	Bare soil, grass, mesh drift fence
Kimberley Rock Goanna	100.0		Mix of front facing (angled orientation) and downward facing	Rocky slab, rocks, leaf litter
Kimberley Rock-rat	100.0	0.0	Mostly front facing, angled orientation	Rocky slab, rocks, leaf litter
Kimberley Ta-ta Lizard	100.0		Mostly front facing, angled orientation, some downward facing	Rocky slab, bare ground, leaf litter
Koala	62.5	37.5	Mostly front facing, perpendicular to ground	Leaf litter, grass, shrubbery, bait canister and pole occasionally present
Kookaburra sp	99.6	0.4	Mostly front facing, angled orientation	Leaf litter, grass, bare soil
Lace Goanna	100.0		Mix of front facing (perpendicular and angled orientation)	Leaf litter, grass, bare soil, some images contain green water trough
Long-haired Rat	89.5	10.5	Mostly front facing, perpendicular to ground	Bare soil, some shrubbery
Long-nosed Bandicoot sp	98.9	1.1	Mostly front facing, angled orientation	Grass, leaf litter, bait canister and pole occasionally present
Long-nosed Potoroo	28.9	71.1	Mostly front facing, perpendicular to ground	Grass, shrubbery, bait canister and pole occasionally present
Magpie-lark	100.0		Mostly front facing, angled orientation	Bare soil, grass, leaf litter
Malleefowl	85.8	14.2	Mostly front facing, perpendicular to ground, some angled orientation	Bare soil, leaf litter, shrubbery

Mareeba Rock-wallaby	100.0		Mostly front facing, angled orientation	Bare soil, leaf litter, grass, bait canister occasionally present
Mertens' Water Goanna	100.0		Mix of front facing (angled orientation) and downward facing	Leaf litter, rocky slab, rocks
Miner Bird sp	99.2	0.8	Mostly front facing, perpendicular to ground	Bare soil (mostly red), grass, some images contain green water trough
Monjon	99.9	0.1	Mostly front facing, angled orientation	Rocky slab, rocks, leaf litter
Mount Claro Rock-wallaby	100.0		Mostly front facing, angled orientation	Rocks, grass, leaf litter, bait canister occasionally present
Musky Rat-kangaroo	100.0		Mostly front facing, perpendicular to ground	Leaf litter, bait canister and pole occasionally present
Nabarlek	100.0		Mostly front facing, angled orientation	Rocky slab, rocks, bare soil, leaf litter
Nocturnal Predatory Bird sp	42.4	57.6	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, some images containing green water trough
Northern Bettong	84.0	16.0	Front facing, angled at 30 from pole	Brown leaf litter, bait canister occasionally present
Northern Nail-tailed Wallaby	94.8	5.2	Mix of front facing (perpendicular and angled orientation)	Grass, bare soil, leaf litter
Numbat	99.0	1.1	Mix of front facing (perpendicular) and downward facing	Leaf litter, bare soil, logs, wooden checkered board
Pale Field-rat	99.8	0.2	Mostly front facing, angled orientation	Bare soil, grass, rocks
Parrot sp	99.4	0.6	Mix of front facing (perpendicular and angled orientation)	Grass, bare soil, shrubbery, some image contain green water trough
Peaceful Dove	100.0		Mostly front facing, angled orientation	Leaf litter, grass, bare soil
Pheasant Coucal	99.6	0.4	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare ground, leaf litter, grass
Pig	95.7	4.3	Mix of front facing (perpendicular and angled orientation)	Grass, leaf litter, trees, bait canister and pole occasionally present
Pseudantechinus sp	50.9	49.1	Mix of front facing (angled orientation) and downward facing	Rocky slab, rocks, bare soil
Pygmy Possum sp	9.1	90.9	Mostly front facing, perpendicular to ground, 5-20cm off ground	Leaf litter, grass, mesh drift fence
Quail sp	98.3	1.8	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil, leaf litter, grass
Quoll sp	78.5	21.6	Mix of front facing (perpendicular and angled orientation) and downward facing	Rocky slab, bare ground, leaf litter, large log, bait canister and star picket occasionally present
Red Fox	35.1	64.9	Mostly front facing, perpendicular to ground	Bare soil, grass, shrubbery, leaf litter
Red Kangaroo	93.9	6.1	Mostly front facing, perpendicular to ground	Bare soil (mostly red), grass, shrubbery, sky

Red-legged Pademelon	98.2	1.8	Mostly front facing, angled orientation	Leaf litter, grass, bait canister occasionally present
Red-necked Wallaby	70.4	29.6	Mix of front facing (perpendicular and angled orientation)	Grass, leaf litter, bare ground, trees, some images contain green water trough
Red-tailed Phascogale	16.3	83.7	Mix of front facing (perpendicular) and downward facing	Bare soil, wooden checkered board, shrubbery, bait canister and pole occasionally present
Ridge-tailed Goanna	100.0		Mix of front facing (angled orientation) and downward facing	Rocky slab, rocks leaf litter, bait canister occasionally present
Ring-tailed Possum sp	49.8	50.2	Mostly front facing, perpendicular to ground	Grass, leaf litter, shrubbery, tree trunk, bait canister and pole occasionally present
Rufous Hare-wallaby	4.4	95.6	Mostly front facing, perpendicular to ground	Bare soil, some shrubbery, black water trough occasionally present
Rufous treecreeper	97.3	2.7	Mostly front facing, perpendicular to ground, some downward facing	Bare soil (mostly red), leaf litter, some wooden logs
Rufus Bettong	100.0		Front facing, angled at 30 from pole	Brown leaf litter, bait canister occasionally present
Scaly-tailed Possum	100.0		Mix of front facing (angled orientation) and upward facing tree trunk	Rocky slab, tree trunk, rocks, leaf litter
Shark Bay Bandicoot	54.5	45.5	Mix of front facing (perpendicular) and downward facing	Bare soil, wooden checkered board
Short-beaked Echidna	77.3	22.7	Mix of front facing (perpendicular and angled orientation) some downward facing	Grass, bare soil, leaf litter, wooden checkered board, shrubbery
Short-nosed Bandicoot sp	74.0	26.0	Mix of front facing (perpendicular and angled orientation) and downward facing	Grass, bare soil, leaf litter, wooden checkered board, bait canister occasionally present
Skink sp	99.3	0.7	Mix of front facing (angled orientation) and downward facing	Rocks, bare soil, leaf litter, rocky slab, wooden checkered board
Small Bird sp	100.0	0.0	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, grass
Spectacled Hare-wallaby	100.0		Mostly front facing, angled orientation	Bare soil (mostly red), grass, leaf litter
Spinifex Pigeon	99.9	0.1	Mix of front facing (perpendicular and angled orientation)	Bare soil (mostly red), rocks, dry grass
Spiny-cheeked Honeyeater	99.9	0.1	Mostly front facing, perpendicular to ground	Lots of red soil, many images containing green water trough and/or perching stick
Swamp Antechinus	86.4	13.6	Front facing, perpendicular to ground	Grass, bait canister and/or wooden pole occasionally present
Swamp Buffalo	98.0	2.0	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, grass (dead and live), trees, sky
Swamp Wallaby	83.0	17.0	Mostly front facing, perpendicular to ground	Leaf litter, grass, bare soil, some images contain green water trough

Tammar Wallaby	56.5	43.5	Mix of front facing (perpendicular) and downward facing	Leaf litter, grass, black solid drift fence, bait canister and pole occasionally present
Water Rat	88.8	11.2	Mostly front facing, angled orientation, some downward facing	Leaf litter, bare soil, rocks
Western Partridge Pigeon	100.0		Mostly front facing, angled orientation	Leaf litter, bare ground, rocks
Western Short-eared Rock-wallaby	98.7	1.3	Mostly front facing, angled orientation	Rocky slab, rocks, bare soil, leaf litter
Wetland Bird sp	99.3	0.8	Mix of front facing (perpendicular and angled orientation)	Leaf litter, trees, grass, water
Whip-tailed Wallaby	100.0		Mix of front facing (perpendicular and angled orientation)	Grass, bare soil, leaf litter, trees
White-quilled Rock-pigeon	100.0		Mix of front facing (angled orientation) and downward facing	Bare soil, rocky slab, leaf litter
White-winged Chough	98.8	1.2	Mix of front facing (perpendicular and angled orientation)	Grass, leaf litter, bare soil
Willie Wagtail	98.2	1.8	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, grass, wooden log, rocks
Wombat sp	9.2	90.8	Mostly front facing, perpendicular to ground	Bare soil, grass, shrubbery, sky
Wren sp	99.4	0.6	Mix of front facing (perpendicular and angled orientation)	Bare soil, leaf litter, some shrubbery
Yellow-footed Antechinus	49.5	50.5	Mix of front facing (perpendicular) and downward facing	Brown leaf litter, grass, wooden checkered board, bait canister and/or wooden pole occasionally present
Yellow-spotted and Sand Goanna sp	99.9	0.1	Mix of front facing (perpendicular and angled orientation) and downward facing	Bare soil (mostly red), wooden checkered board, leaf litter, grass, rocks